Tanmay Bhuskute

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SUMMARY

Software Engineer and M.S. Computer Science candidate (Dec 2026) with 2+ years building production cloud infrastructure, full stack applications, backend automation, and scalable systems using Python, React, Node.js, Docker, Kubernetes, AWS, and Google Cloud Platform.

EDUCATION

Master of Science in Computer Science Aug 2024 to Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Database Theory, Intelligent Data Analysis.

Bachelor of Engineering in Computer Engineering Aug 2018 to Jul 2022
AISSMS College of Engineering
Relevant Coursework: Object Oriented Programming, Data Mining, Machine Learning.

TECHNICAL SKILLS

Programming: Python, Java, JavaScript, TypeScript, SQL, Go, C/C++, Bash.
Web & Backend: React, Node.js, FastAPI, Flask, Express.js, REST APIs, Microservices, API Design.
Databases: PostgreSQL, MySQL, MongoDB, Elasticsearch, Redis, Vector Databases.
Developer Tools: Git, GitLab CI/CD, Jenkins, Docker, Kubernetes, ArgoCD, Helm, Terraform, Linux, Bash.
Cloud & AI: AWS, Google Cloud Platform, Azure, Gemini, Vertex AI, PyTorch, scikit-learn, LangChain, pandas.

WORK EXPERIENCE

Graduate Researcher | University of Cincinnati Feb 2026 to Present

- Designed memory augmented LLM pipeline separating video understanding from automation across **53 runs** on **20+ apps**.
- Reduced average run latency **25.7x** while preserving task completion fidelity through structured evaluation artifacts.

Software Development Engineer | Siemens Industry Software Jul 2022 to Jun 2024

- Automated **Teamcenter X** infrastructure workflows with Python and Boto3, cutting deployment time **40%** across AWS environments.
- Executed **44 production upgrades** in 4 months, debugging Linux, EC2, database, and service issues in live environments.
- Designed Ansible Blue Green deployment framework with rollback support, reducing production upgrade downtime by **50%**.
- Built self service developer platform for environment lifecycle management, adopted by **15 engineers** for daily workflows.
- Migrated **40+ production environments** to Kubernetes using ArgoCD, Helm, GitLab CI/CD, Amazon EFS, and Vault.
- Increased automated code coverage from **35% to 78%**; mentored **5 engineers** on standards and review practices.

Software Developer Intern | Siemens Industry Software Mar 2022 to Jun 2022

- Automated AWS provisioning with Terraform and Go Terratest, cutting deployment time **30%** and catching **95%** misconfigurations.

Software Developer Intern | Government of Maharashtra, Water Resource Department Sep 2021 to Dec 2021

- Built React and MySQL workflow app digitizing record management and improving departmental operating efficiency by **40%**.

PROJECTS AND RESEARCH

PolicyPilot Medical Benefit Policy Intelligence Portal [React, TS, Node, Docker, Gemini] May 2026

- Built React and Express healthcare app unifying prior authorization workflows with REST APIs and structured AI tool calls.
- Designed **17 tool MCP server** with Zod validation for coverage search, patient readiness, and cross payer comparison.

NYC Mobility Forecasting [Python, FastAPI, React, PostgreSQL, Docker] Jul 2026

- Built FastAPI prediction service serving demand forecasts to a React map dashboard with PostgreSQL and Redis caching.
- Modeled demand across **263 zones**, improving MAPE from **18.77% to 16.52%** and cutting RMSE **43%**.

Spotify Personal Analytics Platform [React, TS, Python, FastAPI, PostgreSQL] Jul 2026

- Built React and FastAPI analytics app processing **71K+** Spotify events through Python ETL workflows.
- Designed PostgreSQL schema with materialized views, indexes, and **16 SQL RPC functions** for low latency queries.

EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).